

ISE 5406: Optimization II

Lecture 5

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Reading for this lecture

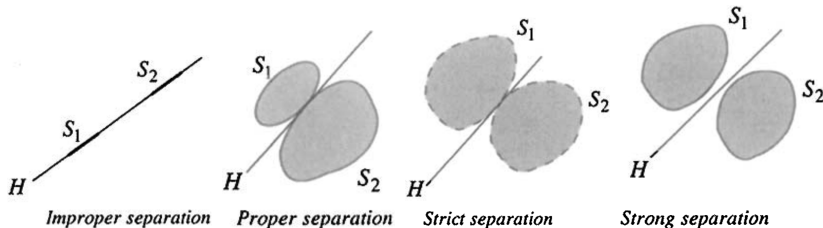
- Sections 2.4, 3.2 and 3.4 of Bazaraa et al. (2006)
- Sections 2.5, 3.1 and 3.2 of Boyd and Vandenberghe

Outline

Recap: Separating Hyperplanes

Let S_1 and S_2 be nonempty sets in $\mathbb{R}^n$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T \mathbf{x} = \alpha\}$ **separates** S_1 and S_2 if $\mathbf{p}^T \mathbf{x} \geq \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T \mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.
- If, additionally, $S_1 \cup S_2 \not\subset H$, then H **properly separates** S_1 and S_2 .
- The hyperplane H **strictly separates** S_1 and S_2 if $\mathbf{p}^T \mathbf{x} > \alpha$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T \mathbf{x} < \alpha$ for each $\mathbf{x} \in S_2$.
- The hyperplane H **strongly separates** S_1 and S_2 if for some $\epsilon > 0$, $\mathbf{p}^T \mathbf{x} \geq \alpha + \epsilon$ for each $\mathbf{x} \in S_1$ and $\mathbf{p}^T \mathbf{x} \leq \alpha$ for each $\mathbf{x} \in S_2$.



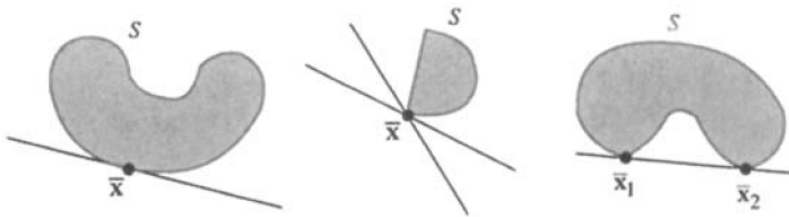
Recap: Supporting Hyperplanes

Let S be a nonempty set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$.

- A hyperplane $H := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) = 0\}$ is called a **supporting hyperplane** of S at $\bar{\mathbf{x}}$ if S is *entirely* contained in one of the halfspaces

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0\}, \quad \text{or} \quad \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0\}$$

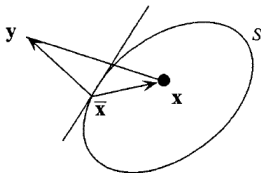
- If $S \not\subseteq H$, then H is a **proper** supporting hyperplane of S at $\bar{\mathbf{x}}$.



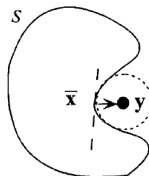
Closest-Point Theorem

Given a nonempty closed convex set $S \subseteq \mathbb{R}^n$ and a point $\mathbf{y} \in \mathbb{R}^n$ with $\mathbf{y} \notin S$, there exists a **unique point $\bar{\mathbf{x}} \in S$ with minimum distance from $\mathbf{y}$** . Moreover, $\bar{\mathbf{x}}$ is the minimizing point if and only if

$$(\mathbf{y} - \bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \leq 0, \text{ for all } \mathbf{x} \in S.$$



(a)



(b)

Separating Hyperplane Theorem

Theorem (Separation of a Convex Set and a Point)

Let S be a nonempty closed convex set in $\mathbb{R}^n$ and $\mathbf{y} \notin S$. Then, there exists a nonzero vector $\mathbf{p}$ and a scalar α such that

$$\mathbf{p}^T \mathbf{y} > \alpha \text{ and } \mathbf{p}^T \mathbf{x} \leq \alpha \text{ for each } \mathbf{x} \in S.$$

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Corollaries

- 1 Let S be a closed convex set in $\mathbb{R}^n$. Then S is the intersection of all halfspaces containing S .
- 2 Let S be a nonempty set, and let $\mathbf{y} \notin \text{cl}(\text{conv}(S))$. Then there exists a strongly separating hyperplane separating S and $\{\mathbf{y}\}$.

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Theorems 2.4.8 and 2.4.10 in Bazaraa et al. present **sufficient conditions for (strong) separation** of two convex sets.

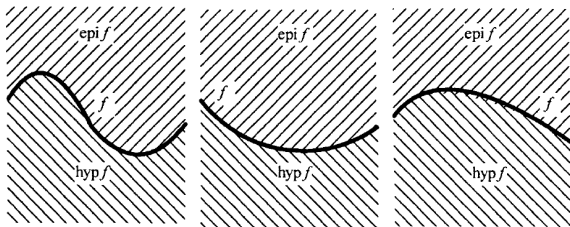
Supporting Hyperplane Theorem

Theorem (Supporting Hyperplane)

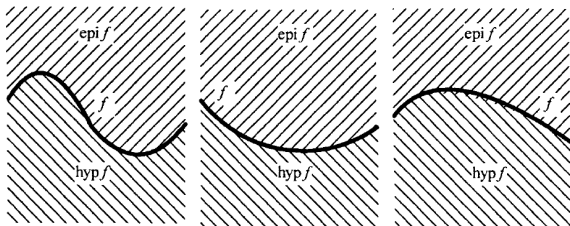
Let S be a nonempty convex set in $\mathbb{R}^n$ and $\bar{\mathbf{x}} \in \partial S$. Then there exists a hyperplane that supports S at $\bar{\mathbf{x}}$. That is, there exists $\mathbf{p} \neq \mathbf{0}$ such that

$$\mathbf{p}^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0 \text{ for each } \mathbf{x} \in \text{cl}(S).$$

Theorem: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. Then f is a convex function if and only if $\text{epi}(f)$ is a convex set.



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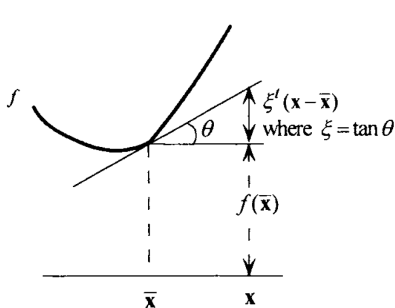


Definition: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be a convex function. Then ξ is called a **subgradient** of f at $\bar{x} \in S$ if

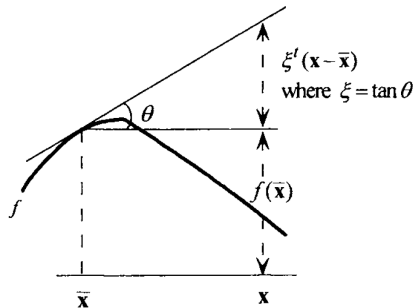
$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \xi^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for all } \mathbf{x} \in S.$$

Definition: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be a **concave** function. Then ξ is called a **subgradient** of f at $\bar{x} \in S$ if

$$f(\mathbf{x}) \leq f(\bar{\mathbf{x}}) + \xi^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for all } \mathbf{x} \in S.$$



Convex function



Concave function

- The affine function $f(\bar{x}) + \xi^T(x - \bar{x})$ corresponds to a supporting hyperplane of the epigraph or the hypograph of the function f depending on whether it is convex or concave.
- The subgradient vector ξ corresponds to the slope of the supporting hyperplane.

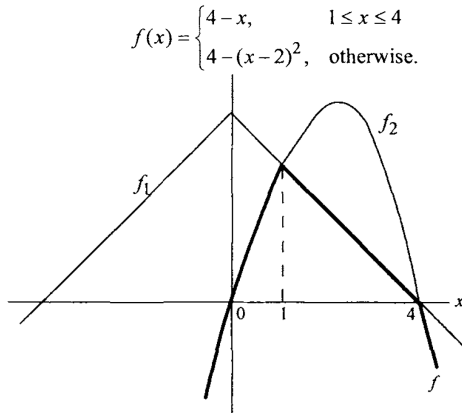
3.2.4 Example

Let $f(x) = \min \{f_1(x), f_2(x)\}$, where f_1 and f_2 are as defined below:

$$f_1(x) = 4 - |x|, \quad x \in \mathbb{R}$$

$$f_2(x) = 4 - (x-2)^2, \quad x \in \mathbb{R}.$$

Since $f_2(x) \geq f_1(x)$ for $1 \leq x \leq 4$, f can be represented as follows:



Theorem

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be a convex function. Then for $\bar{\mathbf{x}} \in \text{int}(S)$, there exists a vector $\boldsymbol{\xi}$ such that the hyperplane

$$H = \{(\mathbf{x}, y) : y = f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})\}$$

supports $\text{epi}(f)$ at $[\bar{\mathbf{x}}, f(\bar{\mathbf{x}})]$. In particular, $\boldsymbol{\xi}$ is a subgradient of f at $\bar{\mathbf{x}}$.

A convex function has a subgradient at every point in the interior of its domain.

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Proof: reading assignment (see Theorem 3.2.5 of Bazaraa et al.)

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Proof: reading assignment (see Theorem 3.2.5 of Bazaraa et al.)

- If f is differentiable at $\bar{\mathbf{x}}$, then $\nabla f(\bar{\mathbf{x}})$ is the *only* subgradient of f at $\bar{\mathbf{x}}$.
- Set of all subgradients of f at $\bar{\mathbf{x}}$ is called the subdifferential $\partial f(\bar{\mathbf{x}}) \subseteq \mathbb{R}^n$.
- $\partial f(\bar{\mathbf{x}})$ is a compact convex set at interior points $\bar{\mathbf{x}}$.

Theorem

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. Suppose for each point $\bar{\mathbf{x}} \in \text{int}(S)$, there exists a subgradient vector $\boldsymbol{\xi}$ such that

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S.$$

Then f is convex on $\text{int}(S)$.

Proof:

Minimizing a Convex Function

Definition: Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a **convex** function and S be a nonempty **convex** set in $\mathbb{R}^n$. Then a convex optimization problem is defined by

$$\begin{array}{ll} \min & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in S. \end{array} \tag{1}$$

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Theorem

Suppose that $\bar{\mathbf{x}} \in S$ is a local optimal solution to Problem (??).

- 1 Then $\bar{\mathbf{x}}$ is a global optimal solution. **[Proof?]**
- 2 If either $\bar{\mathbf{x}}$ is a strict local minimum or f is strictly convex, then $\bar{\mathbf{x}}$ is the unique global optimal solution (it is also a strong local minimum).

Necessary and Sufficient Conditions for a Global Minimum

Theorem

Point $\bar{\mathbf{x}} \in S$ is an optimal solution to Problem (??) **if and only if** f has a subgradient ξ at $\bar{\mathbf{x}}$ such that $\xi^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$.

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Corollary 1: Under the assumptions of the above theorem, if S is an open set (for example, $S = \mathbb{R}^n$), $\bar{\mathbf{x}}$ is an optimal solution to the problem if and only if there exists a **zero subgradient** of f at $\bar{\mathbf{x}}$.

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Corollary 1: Under the assumptions of the above theorem, if S is an open set (for example, $S = \mathbb{R}^n$), $\bar{\mathbf{x}}$ is an optimal solution to the problem if and only if there exists a **zero subgradient** of f at $\bar{\mathbf{x}}$.

Corollary 2: Assume that f is differentiable. Then $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$. Furthermore, if S is open, $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}}) = \mathbf{0}$.